

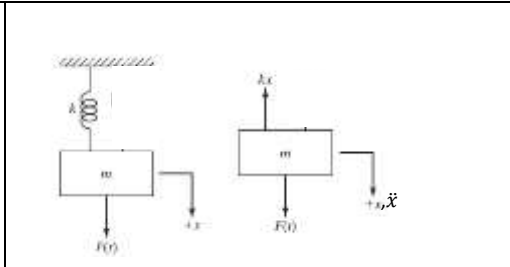
Harmonically Excited Vibration of undamped system

Consider a single degree of freedom spring-mass system subjected to a harmonic force

$$F(t) = F_0 \sin \omega t$$

From the freebody diagram, we can write the equation of motion as

$$m\ddot{x} = F_0 \sin \omega t - kx \quad \Leftrightarrow \quad m\ddot{x} + kx = F_0 \sin \omega t$$



The governing equation is a second order, non homogeneous, linear, ordinary differential equation with constant coefficients. The solution will have two parts; a homogenous solution and a particular solution. The homogeneous solution is obtained from the homogeneous part $m\ddot{x} + kx = 0$

Homogeneous solution:

$$x_h(t) = A_1 \cos \omega_n t + A_2 \sin \omega_n t = X_0 \sin(\omega_n t + \varphi)$$

Particular solution:

Since the excitation is harmonic with frequency ω , let us assume the particular response also as harmonic with frequency ω .

$$x_p(t) = C_1 \sin \omega t + C_2 \cos \omega t \quad \Leftrightarrow \quad \ddot{x}_p = -\omega^2(C_1 \sin \omega t + C_2 \cos \omega t)$$

Substituting this solution back in the differential equation, we get

$$m\{-\omega^2(C_1 \sin \omega t + C_2 \cos \omega t)\} + k\{C_1 \sin \omega t + C_2 \cos \omega t\} = F_0 \sin \omega t$$

$$\{(k - m\omega^2)C_1\} \sin \omega t + \{(k - m\omega^2)C_2\} \cos \omega t = F_0 \sin \omega t$$

Equating the coefficients of $\sin \omega t$ and $\cos \omega t$ on both sides, we get

$$(k - m\omega^2)C_1 = F_0 \quad \Leftrightarrow \quad (k - m\omega^2)C_2 = 0$$

Solving these two equations, the two constants.

$$C_1 = \frac{F_0}{(k - m\omega^2)} \quad \Leftrightarrow \quad C_2 = 0$$

$$\therefore x_p(t) = C_1 \sin \omega t + C_2 \cos \omega t = \frac{F_0}{(k - m\omega^2)} \sin \omega t$$

The complete solution is

$$x(t) = x_h(t) + x_p(t) = A_1 \cos \omega_n t + A_2 \sin \omega_n t + \frac{F_0}{(k - m\omega^2)} \sin \omega t = X_0 \sin(\omega_n t + \varphi) + \frac{F_0}{(k - m\omega^2)} \sin \omega t$$

Using the initial conditions, $x(t=0) = x_0$, $\dot{x}(t=0) = \dot{x}_0$

$$x(t) = A_1 \cos \omega_n t + A_2 \sin \omega_n t + \frac{F_0}{(k - m\omega^2)} \sin \omega t \quad \Leftrightarrow \quad \dot{x}(t) = -A_1 \omega_n \sin \omega_n t + A_2 \omega_n \cos \omega_n t + \frac{F_0}{(k - m\omega^2)} \omega \cos \omega t$$

$$x_0 = A_1 \quad \Leftrightarrow \quad \dot{x}_0 = A_2 \omega_n + \frac{F_0}{(k - m\omega^2)} \omega \quad \Leftrightarrow \quad A_2 = \frac{\dot{x}_0}{\omega_n} - \frac{F_0 \omega}{(k - m\omega^2) \omega_n}$$

$$x(t) = x_0 \cos \omega_n t + \left(\frac{\dot{x}_0}{\omega_n} - \frac{F_0 \omega}{(k - m\omega^2) \omega_n} \right) \sin \omega_n t + \frac{F_0}{(k - m\omega^2)} \sin \omega t$$

Note that the arbitrary constants appear only in the homogeneous part. The homogeneous solution in the case of real systems will eventually become zero because of the damping of the system (however small it may be) as time elapses. Hence, it is called transients. The particular solution will exist as long as the excitation is present. So, it is called the steady-state solution.

The steady state solution

$$x(t)_{steady\ state} = \frac{F_0}{(k - m\omega^2)} \sin \omega t = \frac{\frac{F_0}{k}}{(1 - \frac{m\omega^2}{k})} \sin \omega t = \frac{\frac{F_0}{k}}{(1 - \frac{\omega^2}{\omega_n^2})} \sin \omega t = \frac{\frac{F_0}{k}}{(1 - r^2)} \sin \omega t$$

Where, $r = \frac{\omega}{\omega_n}$ is known as the frequency ratio.

The amplitude of steady state vibration,

$$X = \frac{\frac{F_0}{k}}{(1 - r^2)}$$

$\frac{F_0}{k}$ is the static displacement of the system, if a static force F_0 is acting on the system, where as X is the dynamic displacement amplitude when a dynamic force of magnitude F_0 acts.

$$\frac{X}{\left(\frac{F_0}{k}\right)} = \frac{1}{1 - r^2}$$

This is known as the **amplification factor** or **magnitude ratio** or **amplitude ratio** (β).

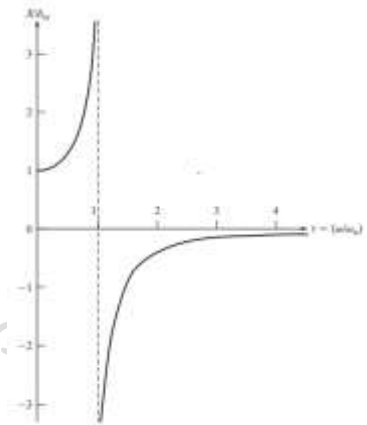
The figure shows amplification factor plotted against frequency ratio. It can be seen that if $r < 1$, the response is positive and the output response is in phase with the input force. When $r > 1$, the response becomes out of phase with the input.

When $r = 1$, or $\omega \rightarrow \omega_n$

$$x(t) = x_0 \cos \omega_n t + \left(\frac{\dot{x}_0}{\omega_n} - \frac{F_0 \omega}{(k - m\omega^2)\omega_n} \right) \sin \omega_n t + \frac{F_0}{(k - m\omega^2)} \sin \omega t$$

$$x(t) = x_0 \cos \omega_n t + \left(\frac{\dot{x}_0}{\omega_n} \right) \sin \omega_n t + \frac{F_0}{(k - m\omega^2)} \sin \omega t - \left(\frac{\omega F_0}{(k - m\omega^2)\omega_n} \right) \sin \omega_n t$$

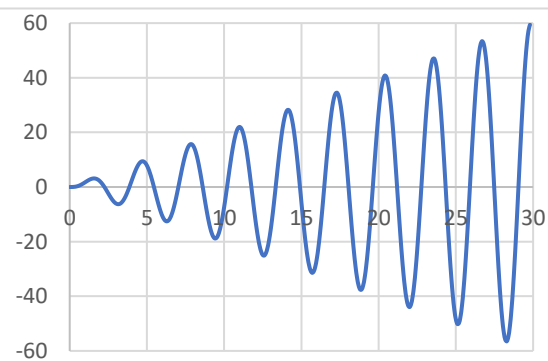
$$x(t) = x_0 \cos \omega_n t + \left(\frac{\dot{x}_0}{\omega} \right) \sin \omega_n t + \frac{F_0}{k} \left\{ \frac{\sin \omega t - \left(\frac{\omega}{\omega_n} \right) \sin \omega_n t}{1 - \left(\frac{\omega}{\omega_n} \right)^2} \right\}$$



When $\omega \rightarrow \omega_n$ or $r \rightarrow 1$, the last term becomes $\frac{0}{0}$ form. Applying L'Hospital's rule,

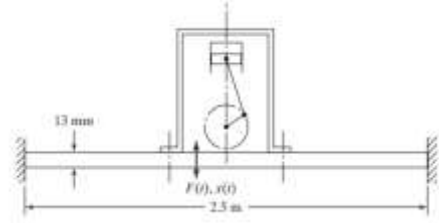
$$\begin{aligned} \lim_{\omega \rightarrow \omega_n} \frac{F_0}{k} \left\{ \frac{\sin \omega t - \left(\frac{\omega}{\omega_n} \right) \sin \omega_n t}{1 - \left(\frac{\omega}{\omega_n} \right)^2} \right\} &= \frac{F_0}{k} \lim_{\omega \rightarrow \omega_n} \left\{ \frac{\sin \omega t - \left(\frac{\omega}{\omega_n} \right) \sin \omega_n t}{1 - \left(\frac{\omega}{\omega_n} \right)^2} \right\} \\ &= \frac{F_0}{k} \lim_{\omega \rightarrow \omega_n} \left\{ \frac{\frac{d}{d\omega} (\sin \omega t - \left(\frac{\omega}{\omega_n} \right) \sin \omega_n t)}{\frac{d}{d\omega} (1 - \left(\frac{\omega}{\omega_n} \right)^2)} \right\} \\ &= \frac{F_0}{k} \lim_{\omega \rightarrow \omega_n} \left\{ \frac{t \cos \omega t - \left(\frac{\sin \omega_n t}{\omega_n} \right)}{-2 \frac{\omega}{\omega_n^2}} \right\} = \frac{F_0}{k} \frac{(\sin \omega_n t - \omega_n t \cos \omega_n t)}{2} \end{aligned}$$

The $(\sin \omega_n t - \omega_n t \cos \omega_n t)$ is plotted by assuming $\omega_n = 2$ as shown. It can be noticed that the amplitude is increasing linearly over a period of time at resonance. The amplitude will not become infinitely large instantaneously.



#Example 1

A reciprocating pump, of mass 75 kg , is mounted at the middle of a steel plate of thickness 13 mm , width 500 mm , and length 2.5 m , clamped along the two edges as shown. During operation of the pump, the plate is subjected to a harmonic force, $F(t) = 250 \cos 62.8t \text{ N}$. Find the amplitude of steady state vibration of the plate. Take $E = 200 \text{ GPa}$.



$$m = 75 \text{ kg} \gg l = 2.5 \text{ m} \gg I = \frac{bh^3}{12} = \frac{0.5 \times 0.013^3}{12} = 9.15 \times 10^{-8} \text{ m}^4 \gg F_0 = 250 \text{ N} \gg \omega = 62.8 \text{ rad/s}$$

$$k = \frac{192EI}{l^3} = \frac{192 \times 200 \times 10^9 \times 9.15 \times 10^{-8}}{2.5^3} = 224870 \text{ N/m}$$

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{224870}{75}} = 54.76 \frac{\text{rad}}{\text{s}} \ll \gg r = \frac{\omega}{\omega_n} = 1.147$$

$$\text{Amplitude of steady state response, } X = \frac{\frac{F_0}{k}}{1-r^2} = \frac{250}{224870(1-1.147^2)} = -3.522 \times 10^{-3} \text{ m}$$

Exercise 1.

A spring-mass system, with a spring stiffness of 5000 N/m , is subjected to a harmonic force of magnitude 30 N and frequency 20 Hz . The mass is found to vibrate with an amplitude of 0.2 m . Assuming that vibration starts from rest ($x_0 = \dot{x}_0 = 0$), determine the mass of the system.

Reference:

Mechanical Vibrations, 6/e, Singiresu S. Rao, Pearson Global Edition

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